

PETALNEX PVT. LTD.

Building intelligent business automation

AI AUTOMATION INTERNSHIP PROGRAM

6-Week Practical Learning, Training & Project Plan

Powered by n8n • LLMs • RAG • AI Agents

36 Working Days • 6 Days / Week (Sunday off) • Online / Remote

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University / Institute	Ghulam Ishaq Khan Institute of Engineering Sciences and Technology
Internship Domain	AI Automation
Duration	6 Weeks (36 working days)
Mode	Online / Remote
Organization	Petalnex Pvt. Ltd.
Supervisor / Team Lead	Sohail Anwar

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1. Program Overview

This program gives university interns hands-on experience designing and building AI-powered business automation using **n8n** and modern AI tooling (LLMs, structured output, RAG, vector databases and AI agents). It is deliberately **project-oriented**: interns do not just study concepts — they build, document, test and demonstrate working workflows every single week.

The working model

Learn → Practice → Build → Document → Demonstrate

By the end of the internship, an intern can

- Understand business processes and identify what is worth automating
- Design and build workflows in n8n using triggers, actions, logic and data transformation
- Work with APIs, webhooks and JSON, and write practical JavaScript in the Code node
- Integrate Google Sheets, Gmail, databases and external services
- Integrate LLMs, apply prompt engineering and force reliable structured output
- Build RAG systems with vector databases and tool-using AI agents
- Implement error handling, logging, security and human-in-the-loop controls
- Take a business requirement and deliver a tested, documented, production-style solution

What makes this version different from a basic plan

- 6 days a week (Sunday off) — 36 working days, giving room for dedicated JavaScript, provider-evaluation, multi-tool-agent and production days a 5-day plan skips
- Real, current learning resources for every day (official n8n Academy/docs plus vetted references)
- Free/low-cost tooling chosen so a cohort of 10+ interns can complete everything at near-zero cost
- A Team Lead playbook and cohort structure designed specifically for running 10+ interns at once

2. How the Internship Runs

Schedule & weekly commitment

Interns work **Monday to Saturday** (Sunday off). Interns manage their own time and complete each day's deliverable before the next day. A suggested daily rhythm:

Block	Approx. time	Focus
Learning	2 – 2.5 hrs	Study the day's concepts using the listed resources
Build	2 – 3 hrs	Implement the practical task / assignment
Research & troubleshoot	60 min	Debug, read docs, test the failing component
Document & report	60 min	Update README and submit the Daily Progress Report

The weekly cycle

- Days 1–5 of each week: learn a concept, build a small workflow, document it.
- Day 6 of each week: a consolidating assignment plus a recorded 2–5 minute demo.
- End of week: live demonstration to the Team Lead (see Section 9).

3. Six-Week Map

Week	Theme	Weekly Deliverable
1	Automation foundations & n8n core	Candidate Screening Automation
2	APIs, webhooks, JavaScript & integration	Automated Lead Management System
3	Generative AI, LLMs & prompt engineering	AI Customer Support Triage System
4	RAG, vector databases & AI agents	Company Knowledge AI Assistant
5	Databases, error handling, security & production	AI-Assisted Recruitment Automation
6	Final capstone project	Complete AI Automation solution + presentation

WEEK 1 — AUTOMATION FOUNDATIONS & n8n CORE

Weekly objective Understand what business automation is, get comfortable inside n8n, and build your first working workflows that move, transform and act on data.

BY THE END OF THIS WEEK YOU CAN

- Explain automation, workflows, triggers, actions and the automation model
- Navigate the n8n editor and run/activate workflows
- Read and write JSON and use n8n expressions (\$json)
- Transform, branch and loop over data using core nodes
- Integrate Google Sheets and Gmail into an automation

Day 1 — Onboarding, Automation Fundamentals & Account Setup

Objective Get set up with every tool you will need, understand automation as a business concept, and run your very first n8n workflow.

LEARN

- What is automation, workflow automation and Business Process Automation (BPA)
- Traditional automation vs AI automation; manual vs automated processes
- The core automation model: Trigger → Input → Processing → Decision → Output/Action
- Where automation adds value: Recruitment, Support, Sales, Marketing, Finance, HR, Reporting, Data Processing
- How the internship works: Learn → Practice → Build → Document → Demonstrate
- Complete the Day-1 Account & Tools Setup Checklist (Appendix B) — do not skip this

LEARNING RESOURCES

- n8n – Try it / Cloud sign-up — <https://n8n.io>
- n8n Academy (official free courses & certificates) — <https://docs.n8n.io/courses/>
- n8n Docs – Learning paths — <https://docs.n8n.io/learning-paths/>

PRACTICAL TASK

Two parts today. First, complete the account setup checklist so you are unblocked for the whole program. Second, identify automation opportunities.

- Create/verify all accounts in Appendix B (n8n, GitHub, Google, Gemini API, Groq, Telegram, Postman)
- Identify 5 real manual business processes that could be automated
- For each, document: Process Name, Current Manual Process, Problem, Trigger, Input, Processing Required, Expected Output, Benefits of Automation

DELIVERABLES

- Automation-Opportunities.pdf (5 processes documented)
- Screenshot proving your n8n workspace and GitHub repo are ready

Day 2 — The n8n Editor — Nodes, Triggers & Your First Workflow

Objective Learn the building blocks of n8n and build a scheduled workflow that sends a message automatically.

LEARN

- n8n editor: canvas, nodes, connections, node panel, executions
- Trigger nodes vs action nodes; Manual Trigger and Schedule Trigger

- Credentials: creating, storing and reusing them safely
- Test workflow vs Active workflow; reading execution results

LEARNING RESOURCES

- n8n – Build your first workflow (tutorial) — <https://docs.n8n.io/try-it-out/>
- n8n Docs – Trigger nodes — <https://docs.n8n.io/integrations/builtin/core-nodes/>

PRACTICAL TASK

Build a Daily Automated Reminder that runs every weekday.

Schedule Trigger → Edit Fields (Set) → Send Email / Telegram Message

- Message example: “Good morning – please submit your daily internship progress before end of day.”
- Schedule it for a fixed weekday time and activate it

DELIVERABLES

- Workflow JSON export
- Screenshot of a successful execution
- 3–4 line explanation of what each node does

Day 3 — Data, JSON & Expressions

Objective Master the data format that flows through every workflow and learn to reference and shape it with expressions.

LEARN

- JSON: key/value pairs, objects, arrays, strings, numbers, booleans, nested values
- n8n data model: items, \$json, referencing another node’s output
- Expressions and the expression editor; template syntax {{ }}
- Data-shaping nodes: Edit Fields (Set)

LEARNING RESOURCES

- MDN – Working with JSON — <https://developer.mozilla.org/en-US/docs/Learn/JavaScript/Objects/JSON>
- n8n Docs – Expressions — <https://docs.n8n.io/code/expressions/>

PRACTICAL TASK

Create student/employee records and classify them automatically.

- Each record has: Name, Email, Department, Score
- Classify: Score ≥ 80 → Excellent; 60–79 → Good; < 60 → Needs Improvement
- Output a clean JSON object per record including the classification

DELIVERABLES

- Working workflow JSON
- Sample input + output shown in a screenshot

Day 4 — Logic & Data Transformation

Objective Route and reshape data using branching, filtering, merging and looping — the backbone of every real workflow.

LEARN

- IF node and Switch node for branching
- Filter node for keeping/removing items
- Merge node for combining data streams

- Loop Over Items (batching) and why you loop
- Edit Fields for adding/renaming/removing fields

LEARNING RESOURCES

- n8n Docs – Flow logic (IF, Switch, Merge, Loop) — <https://docs.n8n.io/flow-logic/>

PRACTICAL TASK

Build a small routing workflow over a list of records.

- Take 10 sample records with a numeric score
- Use IF/Switch to route into three groups (Excellent / Good / Needs Improvement)
- Use Edit Fields to attach a “group” and “next_action” field to each

DELIVERABLES

- Workflow JSON
- Screenshot showing the three routed branches

Day 5 — Google Sheets & Email Automation

Objective Connect n8n to Google Sheets and Gmail to read data, process it, write results back and send personalized email.

LEARN

- Google Sheets node: read rows, append rows, update rows, lookups
- Gmail/Email node: sending personalized email using expressions
- Validating and filtering records before acting on them
- Building a small end-to-end “read → decide → update → notify” pipeline

LEARNING RESOURCES

- n8n Docs – Google Sheets node — <https://docs.n8n.io/integrations/builtin/app-nodes/n8n-nodes-base googlesheets/>
- n8n Docs – Gmail node — <https://docs.n8n.io/integrations/builtin/app-nodes/n8n-nodes-base.gmail/>

PRACTICAL TASK

Build an Internship Application Processing workflow.

Google Sheet → Read Applications → Validate Data → Apply Eligibility Conditions → Categorize Candidate → Update Sheet → Send Email

- Categories: Shortlisted / Needs Review / Not Eligible

DELIVERABLES

- Workflow JSON
- Sample sheet
- Screenshots of an update + a sent email

Day 6 — Week 1 Assignment — Candidate Screening Automation

WEEKLY ASSIGNMENT

Objective Consolidate Week 1 into a complete, documented, demonstrable workflow.

LEARN

- Bring together triggers, Sheets/Email, JSON, expressions and branching into one workflow
- Follow the Standard Submission Format (Appendix D) and Workflow Standards (Appendix G)

PRACTICAL TASK

Project: Candidate Screening Automation. Build a dataset with Name, Email, Degree, Skills, Experience, Availability.

Google Sheets → Read Candidate → Validate Required Fields → Apply Screening Rules → Categorize Candidate
→ Update Result → Send Appropriate Email

DELIVERABLES

- Workflow JSON
 - Screenshots
 - Sample input data
 - README (per Appendix D)
 - 2–5 minute recorded demo
-

WEEK 2 — APIs, WEBHOOKS, JAVASCRIPT & SYSTEM INTEGRATION

Weekly objective Learn how software talks to software: consume APIs, receive live events via webhooks, transform data with JavaScript, and integrate multiple systems into one workflow.

BY THE END OF THIS WEEK YOU CAN

- Understand REST APIs, endpoints, methods and status codes
- Call external APIs with the HTTP Request node using different auth types
- Write practical JavaScript in the Code node to transform data
- Receive and validate live data through webhooks
- Design a multi-system integration end to end

Day 7 — API Fundamentals & Postman

Objective Understand how APIs work by inspecting real requests and responses before wiring anything into n8n.

LEARN

- What an API is; REST, endpoint, request, response
- HTTP methods: GET, POST, PUT, PATCH, DELETE
- Status codes: 200, 201, 400, 401, 403, 404, 429, 500
- Headers, query parameters and request bodies

LEARNING RESOURCES

- MDN – An overview of HTTP — <https://developer.mozilla.org/en-US/docs/Web/HTTP/Overview>
- MDN – HTTP response status codes — <https://developer.mozilla.org/en-US/docs/Web/HTTP/Status>
- Postman – Learning Center — <https://learning.postman.com/docs/getting-started/overview/>
- Public APIs list (for practice) — <https://github.com/public-apis/public-apis>

PRACTICAL TASK

Use Postman with a free/public API (e.g. a weather or quotes API) and inspect it.

- Record the request URL, method and headers
- Capture the full JSON response and describe its structure
- Try one request that returns an error code and note why

DELIVERABLES

- API-Exploration.pdf with request/response screenshots and notes

Day 8 — The HTTP Request Node & Authentication

Objective Call any external API from inside n8n and handle the common authentication patterns.

LEARN

- HTTP Request node: GET/POST, headers, query params, request body
- Authentication: API key, Bearer token, Basic auth, and the OAuth concept
- Storing API keys as n8n credentials (never hard-coded)

LEARNING RESOURCES

- n8n Docs – HTTP Request node — <https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.httprequest/>
- MDN – Using the Fetch API (concept reference) — https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API/Using_Fetch

PRACTICAL TASK

Build a Weather Notification Automation.

Schedule Trigger → Weather API (HTTP Request) → Extract Required Data → Condition (IF) → Send Notification

- Example alert: “Rain is expected today – please carry an umbrella.”

DELIVERABLES

- Workflow JSON
- Screenshot of a notification
- Note on which auth method the API used

Day 9 — JavaScript for Automation (the Code Node)

Objective Learn just enough practical JavaScript to transform data that nodes alone can't — a core skill that separates copy-pasters from builders.

LEARN

- const/let, strings, numbers, objects, arrays, template literals
- if/else, functions, optional chaining (?.)
- Array methods you will use daily: .map(), .filter(), .find(), .reduce()
- The Code node: reading items, returning items, run-once vs run-per-item

LEARNING RESOURCES

- n8n Docs – Code node — <https://docs.n8n.io/code/code-node/>
- MDN – JavaScript first steps — https://developer.mozilla.org/en-US/docs/Learn/JavaScript/First_steps
- javascript.info (free tutorial) — <https://javascript.info/>

PRACTICAL TASK

Transform a messy dataset using only the Code node.

- Input: 15 records with inconsistent casing and a numeric score
- Normalize names (trim + title case), lowercase emails
- Add a computed field grade using .map(), then .filter() to keep only grade ≥ B

DELIVERABLES

- Workflow JSON
- The Code node script pasted into the README
- Before/after sample data

Day 10 — Webhooks — Event-Driven Workflows

Objective Trigger workflows the moment something happens elsewhere, instead of polling on a schedule.

LEARN

- What a webhook is; API vs webhook; polling vs event-driven
- GET vs POST webhooks; Test URL vs Production URL
- Reading the request body; returning a webhook response
- Basic webhook security (secret path, header check)

LEARNING RESOURCES

- n8n Docs – Webhook node — <https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.webhook/>

PRACTICAL TASK

Build a form-submission intake.

Website/Form → n8n Webhook → Validate Submitted Information → Save Data → Send Confirmation

- Test it by posting sample JSON from Postman to your webhook URL

DELIVERABLES

- Workflow JSON
- Sample POST request from Postman
- Screenshot of a saved record + confirmation

Day 11 — Multi-System Integration

Objective Combine a webhook, an external API, storage and a notification channel into one coherent pipeline.

LEARN

- Sequencing multiple systems reliably
- Enriching incoming data with an API lookup
- Choosing where to store data and how to notify a human/team

PRACTICAL TASK

Build a Lead Processing Workflow using at least one webhook, one external API, one data store and one notification channel.

New Lead → Webhook → Validate → API Lookup / Enrich → Save Lead → Notify Sales Team

DELIVERABLES

- Workflow JSON
- README
- Screenshots of each stage working

Day 12 — Week 2 Assignment — Automated Lead Management System

WEEKLY ASSIGNMENT

Objective Deliver a complete, scored, routed lead-management automation.

PRACTICAL TASK

Project: Automated Lead Management System. Accept leads with fields: name, email, company, service, budget.

Webhook → Validation → Lead Processing → Lead Scoring → IF / Switch → Route

- High Priority → immediate sales notification
- Medium Priority → save to lead database/sheet
- Low Priority → add to follow-up / nurture list
- Finally: send an automatic acknowledgment email

DELIVERABLES

- Workflow JSON
- README
- Sample API request
- Screenshots
- 2–5 min demo

WEEK 3 — GENERATIVE AI, LLMs & PROMPT ENGINEERING

Weekly objective Bring intelligence into your workflows: understand how LLMs work, write disciplined prompts, integrate models into n8n, and force reliable structured output.

BY THE END OF THIS WEEK YOU CAN

- Explain LLM fundamentals: tokens, context, hallucination, cost, limitations
 - Write structured prompts using a repeatable framework
 - Call an LLM inside n8n and route on its output
 - Produce and validate predictable JSON output from a model
 - Compare providers and evaluate prompt quality with test cases
-

Day 13 — Generative AI & LLM Fundamentals

Objective Understand what large language models can and cannot do, and set up free/low-cost model access for the rest of the program.

LEARN

- AI vs ML vs Generative AI vs LLMs (conceptual)
- Tokens, context window, system instructions, user input
- Hallucination, model limitations and cost awareness
- Providers: OpenAI, Google Gemini, Anthropic Claude, and local models via Ollama
- Set up free model access: Google AI Studio (Gemini) and Groq keys (see Appendix B/C)

LEARNING RESOURCES

- n8n Docs – Advanced AI / LangChain overview — <https://docs.n8n.io/advanced-ai/>
- Google AI Studio (free Gemini API key) — <https://aistudio.google.com/>
- Groq Console (free, very fast inference) — <https://console.groq.com/>
- Ollama (run models locally, free) — <https://ollama.com/>

PRACTICAL TASK

Prove your model access works.

- Generate a Gemini API key and a Groq API key
- Make one successful test call to each from Postman
- Write a half-page note: which you'd use for classification vs long-context work, and why

DELIVERABLES

- Screenshots of successful test calls (keys redacted)
 - Your provider comparison note
-

Day 14 — Prompt Engineering

Objective Write prompts that are specific, constrained and reliable enough to drive automation logic.

LEARN

- The prompt framework: ROLE → CONTEXT → TASK → RULES → OUTPUT FORMAT → EXAMPLES
- Why “Analyze this lead” fails and a structured prompt succeeds
- Few-shot examples and explicit output contracts

LEARNING RESOURCES

- Google – Prompting guide (Gemini) — <https://ai.google.dev/gemini-api/docs/prompting-strategies>
-

- Anthropic – Prompt engineering overview — <https://docs.claude.com/en/docs/build-with-claude/prompt-engineering/overview>

PRACTICAL TASK

Write strong, reusable prompts for five automation tasks.

- Lead qualification, Email classification, CV analysis, Complaint classification, Support reply
- Each prompt must specify role, rules and a strict JSON output shape
- Test each prompt against 3 sample inputs and record the outputs

DELIVERABLES

- Prompt-Library.pdf (5 prompts + test outputs)

Day 15 — LLM Integration inside n8n

Objective Wire a model into a workflow and route downstream logic on what it returns.

LEARN

- Basic LLM Chain vs the AI Agent node — when to use which
- Connecting a chat-model sub-node (Gemini/OpenAI/Groq/Ollama) to an AI node
- Passing dynamic input into the prompt via expressions
- Routing on the model's output with Switch

LEARNING RESOURCES

- n8n Docs – Basic LLM Chain node — <https://docs.n8n.io/integrations/builtin/cluster-nodes/root-nodes/n8n-nodes-langchain.chainllm/>
- n8n Docs – AI/LangChain nodes — <https://docs.n8n.io/integrations/builtin/cluster-nodes/>

PRACTICAL TASK

Build an AI Email Classification workflow.

New Email → AI Analysis → Classify → Route by category

- Categories: Sales, Support, Complaint, Invoice, Spam, General

DELIVERABLES

- Workflow JSON
- The prompt used
- Screenshots of at least 3 different classifications

Day 16 — Structured AI Output

Objective Stop relying on free-form text — make the model return predictable, parseable data your workflow can trust.

LEARN

- Why automation logic needs structured output, not prose
- JSON output, output parsers and schema definition in n8n
- Parsing, validation, handling missing values and malformed responses
- Fallback handling when the model returns something unexpected

LEARNING RESOURCES

- n8n Docs – Structured Output Parser — <https://docs.n8n.io/integrations/builtin/cluster-nodes/sub-nodes/n8n-nodes-langchain.outputparserstructured/>

PRACTICAL TASK

Force a strict JSON contract and validate it.

- Target shape: { category, priority, sentiment, department, summary }
- Add an output parser + a validation/fallback branch for bad responses
- Deliberately feed one confusing input and prove the fallback triggers

DELIVERABLES

- Workflow JSON
- The JSON schema
- Screenshot of the fallback path firing

Day 17 — Provider Comparison & Prompt Evaluation

Objective Choose models deliberately and measure prompt quality instead of guessing.

LEARN

- Comparing Gemini vs Groq (and local Ollama) on the same task: quality, speed, cost, rate limits
- Building a small test set to evaluate a prompt objectively
- Reading token usage and estimating cost per 1,000 runs

LEARNING RESOURCES

- OpenRouter – Free LLM APIs compared — <https://openrouter.ai/blog/tutorials/free-llm-apis-compared/>

PRACTICAL TASK

Evaluate one classification prompt across two providers.

- Run the same 10 test inputs through Gemini and Groq
- Score each output pass/fail against the expected result
- Record accuracy, average latency and estimated cost; recommend one provider

DELIVERABLES

- Evaluation-Report.pdf (test table + recommendation)

Day 18 — Week 3 Assignment — AI Customer Support Triage System

WEEKLY ASSIGNMENT

Objective Deliver a production-style AI triage workflow with validated structured output.

PRACTICAL TASK

Project: AI Customer Support Triage System. Input is a customer support message; the AI must determine category, priority, sentiment, department, summary and a suggested response.

Customer Message → Webhook → AI Analysis → Structured JSON → Switch by Category → Store Ticket → Send Acknowledgment → Urgent Alert if required

DELIVERABLES

- Workflow JSON
- The prompt used
- Structured JSON schema
- At least 10 test cases with results
- README
- 2–5 min demo

WEEK 4 — RAG, VECTOR DATABASES & AI AGENTS

Weekly objective Build AI that uses your organization's own knowledge and can take actions — retrieval-augmented generation and tool-using agents.

BY THE END OF THIS WEEK YOU CAN

- Explain RAG architecture end to end
- Ingest documents into a vector database
- Build a grounded RAG assistant with source-aware, fallback-safe answers
- Build a tool-using AI agent and understand agent vs workflow
- Extend an agent with multiple tools (including MCP)

Day 19 — RAG Fundamentals

Objective Understand how retrieval-augmented generation grounds an LLM in real, specific knowledge.

LEARN

- RAG = Retrieval-Augmented Generation; why it beats a bare LLM for company knowledge
- Ingestion path: Documents → Extract → Chunk → Embed → Store in vector DB
- Query path: Question → Search → Retrieve context → Send to LLM → Grounded answer
- Key ideas: embeddings, similarity search, chunking, metadata, grounding

LEARNING RESOURCES

- n8n Docs – RAG in n8n — <https://docs.n8n.io/advanced-ai/rag-in-n8n/>
- n8n Docs – Vector stores — <https://docs.n8n.io/integrations/builtin/cluster-nodes/sub-nodes/>

PRACTICAL TASK

Write, in your own words, a one-page explanation of RAG.

- Draw the ingestion and query diagrams
- Explain chunking and why chunk size matters
- List 3 real problems in Petalnexus businesses that RAG would solve

DELIVERABLES

- RAG-Explained.pdf (diagrams + explanation)

Day 20 — Document Ingestion & Embeddings

Objective Build the pipeline that turns raw documents into searchable vectors.

LEARN

- Extracting content from Sheets, text, PDF and FAQ files
- Cleaning and chunking content sensibly
- Generating embeddings and storing them with metadata
- Vector store options: Supabase Vector, Qdrant, Pinecone (all have free tiers)

LEARNING RESOURCES

- n8n Docs – Supabase Vector Store — <https://docs.n8n.io/integrations/builtin/cluster-nodes/root-nodes/n8n-nodes-langchain.vectorstoresupabase/>
- Supabase (free Postgres + pgvector) — <https://supabase.com/>
- Qdrant Cloud (free tier) — <https://qdrant.tech/>

PRACTICAL TASK

Build a document-ingestion workflow.

Document → Extract Content → Clean Content → Split into Chunks → Generate Embeddings → Store in Vector DB

- Use at least 10 knowledge entries
- Confirm rows/vectors appear in your vector store

DELIVERABLES

- Workflow JSON
- Screenshot of stored vectors
- The source documents used

Day 21 — Build a RAG Assistant

Objective Answer questions strictly from your own knowledge base, with a safe fallback and source citation.

LEARN

- Wiring a retriever + vector store + chat model for Q&A
- Prompting for grounded answers (“only use provided context”)
- Fallback when the answer isn’t in the knowledge base
- Returning the source/reference for trust

PRACTICAL TASK

Build a Company / HR Policy Assistant.

- Knowledge: leave policy, working hours, attendance, internship guidelines, code of conduct
- Example question: “How many casual leaves are allowed?”
- It must NOT invent policies; unknown → clear “information not available” fallback; mention source where possible

DELIVERABLES

- Workflow JSON
- 5 example Q&A screenshots (including one fallback)
- README

Day 22 — AI Agents — Tools, Memory & Reasoning

Objective Move from fixed workflows to goal-driven agents that choose and call tools.

LEARN

- Standard workflow (fixed sequence) vs AI Agent (goal → reason → choose tool → act → observe)
- The AI Agent node: system prompt, connected tools, memory
- An agent is NOT just a chatbot: it has instructions, tools, context, boundaries and decision capability
- Common agent tools: search knowledge base, read Sheet, query DB, call API, send email, create ticket

LEARNING RESOURCES

- n8n Docs – AI Agent node — <https://docs.n8n.io/integrations/builtin/cluster-nodes/root-nodes/n8n-nodes-langchain.agent/>
- n8n Docs – Memory sub-nodes — <https://docs.n8n.io/integrations/builtin/cluster-nodes/sub-nodes/>

PRACTICAL TASK

Build a single-tool agent, then add memory.

Chat Trigger → AI Agent → Tool: Knowledge Base search → Response

- Give it a strict system prompt (ask when unsure, never invent)

- Add conversation memory and prove it remembers earlier turns

DELIVERABLES

- Workflow JSON
- System prompt
- Chat transcript screenshot showing memory working

Day 23 — Multi-Tool Agents & MCP

Objective Give an agent several tools — including remote tools via MCP — and control which decisions it may make.

LEARN

- Adding multiple tools (HTTP, database query, Google Sheets, Calculator) to one agent
- Writing precise tool descriptions so the agent picks correctly
- MCP Client Tool: letting an agent use tools exposed by a remote MCP server (a mid-2026 n8n capability)
- Boundaries: what the agent may do autonomously vs what needs a human

LEARNING RESOURCES

- n8n Docs – MCP Client Tool — <https://docs.n8n.io/integrations/builtin/cluster-nodes/sub-nodes/n8n-nodes-langchain.toolmcp/>

PRACTICAL TASK

Extend your Day-22 agent with at least two more tools.

- Add an API/HTTP tool and a data-store tool
- Tighten tool descriptions until the agent routes reliably
- Add one boundary: a sensitive action that requires human confirmation

DELIVERABLES

- Workflow JSON
- List of tools + descriptions
- Screenshots of the agent using different tools

Day 24 — Week 4 Assignment — Company Knowledge AI Assistant

WEEKLY ASSIGNMENT

Objective Deliver a complete RAG/agent assistant over real company knowledge.

PRACTICAL TASK

Project: Company Knowledge AI Assistant. Ingest company knowledge into a vector DB, then answer user questions with retrieval + generation.

Company Knowledge → Ingestion → Embeddings → Vector DB → User Question → RAG / Agent → Retrieve → Generate → Return Response

- At least 20 knowledge entries
- Vector search + AI-generated answer + “information not available” fallback + source where possible
- Bonus: conversation memory, human escalation, or a Telegram/chat interface

DELIVERABLES

- Workflow JSON
- Knowledge source files
- 10 example queries with results

- README
 - 2–5 min demo
-

WEEK 5 — DATABASES, ERROR HANDLING, SECURITY & PRODUCTION

Weekly objective Turn demo workflows into reliable, secure, production-grade automations that fail gracefully and keep humans in control of sensitive decisions.

BY THE END OF THIS WEEK YOU CAN

- Model and operate a database (CRUD) from n8n
- Design error handling, retries, logging and alerts
- Apply security and secret-management best practices
- Insert human approval into sensitive AI workflows
- Monitor, debug and optimize workflows for production

Day 25 — Database Fundamentals & CRUD

Objective Store structured data properly in a real database and operate it from n8n.

LEARN

- Database, table, row, column, primary key; CRUD (Create, Read, Update, Delete)
- SELECT / INSERT / UPDATE / DELETE basics
- PostgreSQL and Supabase (managed Postgres, free tier)
- Connecting a Postgres/Supabase node to n8n

LEARNING RESOURCES

- n8n Docs – Postgres node — <https://docs.n8n.io/integrations/builtin/app-nodes/n8n-nodes-base.postgres/>
- Supabase Docs – Getting started — <https://supabase.com/docs>

PRACTICAL TASK

Create a tickets table and operate it from n8n.

- Columns: ticket_id, customer_name, email, category, priority, status, ai_summary, created_at
- Do all four CRUD operations from workflows (insert, read, update status, delete a test row)

DELIVERABLES

- Workflow JSON(s)
- Screenshot of table with rows
- The SQL you used

Day 26 — Error Handling & Reliability

Objective Make workflows survive the real world: bad data, failed APIs, timeouts and rate limits.

LEARN

- Failure modes: workflow errors, API failures, invalid/missing data, auth failures, timeouts, rate limiting (429)
- Retry settings, Error Trigger / error workflows, try-catch patterns
- Error logging and error notifications; fallback actions
- Using execution history to debug

LEARNING RESOURCES

- n8n Docs – Error handling — <https://docs.n8n.io/flow-logic/error-handling/>
- n8n Docs – Executions & debugging — <https://docs.n8n.io/workflows/executions/>

PRACTICAL TASK

Harden an existing workflow.

API Request → Successful? — YES → Continue → NO → Retry → Log Error → Notify Admin

- Add a dedicated error workflow that logs failures and alerts you
- Force a failure and prove the alert fires

DELIVERABLES

- Workflow JSON + error workflow
- Screenshot of a logged/alerted failure

Day 27 — Security & Secret Management

Objective Protect credentials, user data and endpoints — the difference between a toy and a trustworthy system.

LEARN

- API keys, credentials, environment variables, secret management
- Authentication vs authorization; least privilege
- PII and sensitive data handling; webhook security
- Never: hard-code passwords, commit keys to GitHub, share secrets in screenshots, store secrets in docs, or expose open webhooks

LEARNING RESOURCES

- n8n Docs – Credentials — <https://docs.n8n.io/credentials/>
- OWASP – API Security Top 10 — <https://owasp.org/www-project-api-security/>

PRACTICAL TASK

Security-audit your own Week-1-to-Week-4 work.

- Confirm no secrets are hard-coded or committed; move any into credentials/env
- Add a secret/header check to one webhook
- Write a short checklist of what you fixed

DELIVERABLES

- Security-Audit.pdf (findings + fixes)
- Screenshot of the secured webhook

Day 28 — Human-in-the-Loop AI

Objective Keep humans in control of decisions that carry real consequences.

LEARN

- Why AI should not auto-decide hiring, payments, refunds, legal, financial or sensitive customer matters
- Approval patterns: Wait node, approval via email/Telegram, resume-on-response
- Treating AI output as decision support, not the final decision

LEARNING RESOURCES

- n8n Docs – Wait node — <https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base.wait/>

PRACTICAL TASK

Add an approval stage to an existing AI workflow.

AI Recommendation → Human Review (approve/reject) → Approved → Final Action → Rejected → Stop/notify

DELIVERABLES

- Workflow JSON
- Screenshot of the approval step in action

Day 29 — Monitoring, Debugging & Optimization

Objective Operate workflows like a professional: observe them, debug fast and make them efficient.

LEARN

- Reading execution history; pinning data and re-running from a node while debugging
- Logging strategy: what to log and where
- Performance: reducing unnecessary AI calls, batching, caching lookups
- Deployment awareness: n8n Cloud vs self-hosted (Docker + Postgres), activating workflows safely

LEARNING RESOURCES

- n8n Docs – Debug & re-run — <https://docs.n8n.io/workflows/executions/debug/>
- n8n Docs – Hosting/self-host overview — <https://docs.n8n.io/hosting/>

PRACTICAL TASK

Optimize one of your earlier workflows.

- Identify one redundant or expensive step (often an unnecessary AI call) and remove/replace it with deterministic logic
- Add sensible logging
- Document the before/after (steps, runtime, or cost)

DELIVERABLES

- Optimized workflow JSON
- Before/after optimization note

Day 30 — Week 5 Assignment — AI-Assisted Recruitment Automation

WEEKLY ASSIGNMENT

Objective Deliver a secure, error-handled, human-approved recruitment automation.

PRACTICAL TASK

Project: AI-Assisted Recruitment Automation. AI assists; authorized humans make the final call.

Application → Store Candidate Data → Extract Candidate Information → AI Analysis + Skills Extraction → Rule-Based + AI Assessment → Human Review → Shortlist / Reject → Candidate Email → Update Database / Sheet

- Example output: { candidate, skills[], score, recommendation, reason }
- Final hiring/selection decisions must remain with authorized humans

DELIVERABLES

- Full workflow JSON
- AI prompt
- Human approval mechanism
- Error handling
- Documentation (README)
- 2–5 min demo

WEEK 6 — FINAL CAPSTONE PROJECT

Weekly objective Design, build, test, document and present a complete AI Automation solution for a real business problem — the proof that you can take a requirement and deliver a reliable working system.

BY THE END OF THIS WEEK YOU CAN

- Turn a business requirement into a solution design
 - Architect and build an end-to-end automation with AI where appropriate
 - Test it against normal, edge and failure cases
 - Document and present it professionally
-

Day 31 — Requirement Analysis

Objective Choose your capstone and define the problem precisely before building anything.

LEARN

- Writing a clear problem statement and describing the current process
- Identifying pain points: manual work, time, repetition, errors, delays
- Proposing a solution: trigger, input, workflow, AI role, business rules, data storage, output, notifications
- Pick ONE from the Final Project Options (see that section) after discussing with your Team Lead

PRACTICAL TASK

Produce your Project Requirement Document.

- Problem statement
- Current process + pain points
- Proposed solution outline
- Success criteria

DELIVERABLES

- Project-Requirement-Document.pdf
-

Day 32 — Solution Architecture

Objective Design the full system on paper so building is execution, not improvisation.

LEARN

- Mapping the end-to-end flow: trigger → validation → business logic → AI → RAG/API → database → notifications → output
- Documenting workflows, APIs, AI models, database, vector DB, credentials, failure handling, security

PRACTICAL TASK

Produce an architecture diagram and component list.

- Architecture diagram (draw.io/Excalidraw is fine)
- List every node group, integration, model and data store
- Note failure handling and security considerations

DELIVERABLES

- Architecture-Diagram (image/PDF)
 - Component & credentials list
-

Day 33 — Development — Core Build

Objective Build the backbone: trigger, validation, business logic and data storage.

LEARN

- Build the deterministic core first (before AI), following Workflow Standards (Appendix G)
- Professional node naming, clean organization, safe credential use

PRACTICAL TASK

Implement the non-AI core of your capstone.

- Trigger + validation working
- Business rules + storage working
- Clean, named, testable nodes

DELIVERABLES

- Work-in-progress workflow JSON
- Screenshot of the core running

Day 34 — Development — AI, Integration & Notifications

Objective Layer in AI, RAG/agent, external APIs and notifications.

LEARN

- Adding AI only where reasoning/understanding/generation is genuinely needed
- Structured output + validation + fallback in the real pipeline
- Wiring notifications and any human approval

PRACTICAL TASK

Complete the intelligent layer of your capstone.

- AI/RAG/agent integrated with structured output
- External API + notifications wired
- Error handling in place

DELIVERABLES

- Feature-complete workflow JSON

Day 35 — Testing & Optimization

Objective Prove it works — and holds up when things go wrong.

LEARN

- Test categories: normal, missing data, invalid data, external failure, AI failure, database failure, security
- Recording Test Input / Expected / Actual / Pass-Fail and fixing gaps

PRACTICAL TASK

Write and run a full test pass.

- Minimum 15 test cases across all categories
- Fix every failure you find
- Confirm no secrets are exposed

DELIVERABLES

- Test-Cases.pdf (≥15 cases with results)
- Final hardened workflow JSON

Day 36 — Final Presentation & Demo Day

Objective Present your solution professionally and demonstrate it live.

LEARN

- Structuring a clear technical presentation
- Explaining decisions: why these nodes, why AI (or not), how failures are handled, where credentials live

PRACTICAL TASK

Prepare and deliver your final presentation.

- Cover: title, business problem, existing process, proposed solution, architecture, technologies, workflows, AI integration, live demo, error handling, security, business benefits, limitations, future improvements

DELIVERABLES

- Presentation (PDF/slides)
 - Live demonstration
 - Final repository with full documentation
-

Final Capstone — Project Options

Choose ONE option for Week 6 after discussing with your Team Lead. Each must include validation, AI where appropriate, storage, error handling, security and documentation.

Option 1 — AI Customer Support Automation

- Inquiry intake
- AI classification
- Priority detection
- Sentiment analysis
- Knowledge-base search
- Suggested response
- Ticket creation
- Human escalation

Option 2 — AI Recruitment Automation

- Application collection
- Candidate data & skills extraction
- Candidate summarization
- AI-assisted evaluation
- Human review
- Interview workflow
- Automated emails

Option 3 — AI Lead Qualification System

- Lead → validation → enrichment
- AI qualification + lead score
- CRM / database write
- Sales alert
- Follow-up / nurture

Option 4 — Company Knowledge AI Assistant

- Policies, SOPs, FAQs, internal docs
- RAG + embeddings + vector DB
- Source-aware responses
- Unknown-answer fallback
- Human escalation

Option 5 — AI Complaint Management System

- Complaint intake + classification
- Category + priority
- Department routing
- Case creation
- Notification
- Management summary

Option 6 — Automated AI Management Reporting

- Fetch KPIs from DB/spreadsheet
- Calculate metrics

- AI analysis + executive summary
- Risks / insights
- Recommendations
- Email / report delivery

Appendix B — Day-1 Accounts & Tools Setup Checklist

Complete all of these on Day 1. All have free tiers suitable for the whole program.

Tool	Purpose	Link
n8n (Cloud trial or self-host)	Build all workflows	https://n8n.io
GitHub	Store repos, README, workflow JSON	https://github.com
Google account	Google Sheets + Gmail nodes	https://accounts.google.com
Google AI Studio (Gemini)	Free LLM API key	https://aistudio.google.com
Groq Console	Free, very fast LLM inference	https://console.groq.com
Supabase	Free Postgres DB + vector store	https://supabase.com
Telegram	Bot for notifications / chat interface	https://telegram.org
Postman	Test APIs before wiring them	https://www.postman.com
Ollama (optional)	Run local models for free	https://ollama.com

Security note: store every key as an n8n credential or environment variable. Never hard-code keys or commit them to GitHub, and never include them in screenshots.

Appendix C — Core Learning Resource Library

n8n (primary)

- n8n Academy — official free courses & certificates — <https://docs.n8n.io/courses/>
- n8n Learning paths — <https://docs.n8n.io/learning-paths/>
- Build your first workflow — <https://docs.n8n.io/try-it-out/>
- n8n Documentation (nodes, expressions, credentials) — <https://docs.n8n.io/>
- n8n Advanced AI (LLMs, agents, RAG, vector stores) — <https://docs.n8n.io/advanced-ai/>
- n8n Community forum — <https://community.n8n.io/>

JSON, HTTP & JavaScript

- MDN — Working with JSON — <https://developer.mozilla.org/en-US/docs/Learn/JavaScript/Objects/JSON>
- MDN — HTTP overview & status codes — <https://developer.mozilla.org/en-US/docs/Web/HTTP>
- javascript.info — free JS tutorial — <https://javascript.info/>
- Postman Learning Center — <https://learning.postman.com/>

LLMs & prompt engineering

- Google AI Studio (free Gemini key) — <https://aistudio.google.com>
- Groq Console (free, fast) — <https://console.groq.com>
- Google — Gemini prompting strategies — <https://ai.google.dev/gemini-api/docs/prompting-strategies>
- Anthropic — prompt engineering — <https://docs.claude.com/en/docs/build-with-claude/prompt-engineering/overview>
- OpenRouter — free LLM APIs compared — <https://openrouter.ai/blog/tutorials/free-llm-apis-compared/>

Databases, RAG & vector stores

- Supabase docs (Postgres + pgvector) — <https://supabase.com/docs>
- Qdrant (vector DB, free tier) — <https://qdrant.tech/>
- Pinecone (vector DB, free tier) — <https://www.pinecone.io/>

Version control

- GitHub — Hello World / getting started — <https://docs.github.com/en/get-started>

Note: *free-tier limits and model names change often — always defer to the official provider/model docs when implementing a specific integration.*

Appendix D — Standard Submission Format

For every weekly assignment, use this repository structure:

```
project-name/  
|  
├── workflow/  
│   └── workflow.json  
|  
├── screenshots/  
|  
├── sample-data/  
|  
└── README.md
```

README must include

- Project name
- Problem statement
- Objective
- Workflow architecture
- Technologies used
- Nodes used
- Setup instructions
- Credentials required (names only — never values)
- Workflow explanation
- Test cases
- Error handling
- Known limitations
- Future improvements

Never include actual passwords or API secrets in any file, README or screenshot.

Appendix E — Daily Progress Report Template

Submit this every working day (Mon–Sat):

Date:
Today I learned:
Today I built / practiced:
Technologies / nodes used:
Problem(s) faced:
How I solved them:
Task status: Not started / In progress / Completed
GitHub / workflow link:
Plan for next day:

Appendix F — Weekly Demonstration Questions

At each weekly demo, be ready to answer:

- What problem does your workflow solve?
- What triggers it?
- What data does it receive?
- Why did you choose these nodes?
- How does your logic/condition work?
- What happens if an API fails?
- Where are credentials stored?
- Why is AI needed here — and could this be solved without AI?
- How would you improve the workflow?

You must understand your own workflow. Copying a tutorial without understanding it does not count as successful completion.

Appendix G — Workflow Development Standards

Do not leave default node names

Bad: HTTP Request1, IF1, Set2, Code3

Good: Fetch Candidate Records, Validate Required Information, Calculate Eligibility, Generate AI Assessment, Store Candidate Result, Notify HR Team

Name workflows clearly

- HR - Candidate Screening - v1
- Sales - Lead Qualification - v1
- Support - AI Ticket Classification - v1

Keep every workflow

- Organized
- Readable
- Properly named
- Documented
- Secure
- Testable

Appendix H — AI Development Rules

Do not

- Blindly trust AI responses
- Let AI make critical decisions without controls
- Assume AI output is always correct
- Use unstructured AI output when logic depends on predictable data

Do

- Use structured outputs wherever possible
- Validate AI output
- Add fallback logic
- Write clear, constrained prompts
- Test multiple cases
- Keep humans in sensitive decisions

Rule of thumb: use AI where reasoning, understanding, summarization, classification or generation is required. Use rules / IF conditions / Code where deterministic business logic is sufficient. Do not use AI unnecessarily — it adds cost and unpredictability.

Appendix I — Evaluation Rubric

Evaluation Area	Weight
Weekly assignments	25%
Technical understanding	15%
Workflow quality	15%
Final capstone project	20%
Problem solving	10%
Documentation	10%
Participation & professionalism	5%

Result	Threshold	Subject to
Pass	60% overall	Meeting completion requirements (Appendix J)
Outstanding	85% or above	Strong project quality, independent problem solving, good documentation, technical understanding, timely submissions

Appendix J — Completion Requirements & Problem-Solving

To successfully complete the internship

- Maintain required attendance/participation (Mon–Sat)
- Complete assigned learning activities
- Submit weekly projects
- Complete the final capstone project
- Submit workflow source files and final documentation
- Complete the final demonstration/presentation
- Demonstrate genuine understanding of the technologies used

Before asking the Team Lead for help, you must have

- Read the error carefully
- Checked node input and output
- Reviewed execution history
- Verified credentials
- Checked the API/official documentation
- Tested the failing component separately
- Documented what you already tried

When you do ask, provide: the problem, expected result, actual result, error message, a screenshot, and what you already tried.

Expected Final Competency

By the end of this internship, you should be able to receive a business requirement such as:

“We receive customer inquiries through our website. We want to automatically understand each inquiry, categorize it, check our internal knowledge base, store the inquiry, notify the right team, send an acknowledgment, and escalate urgent cases.”

...and then design and deliver:

Requirement Analysis → Process Flow → Workflow Architecture → Webhook / Trigger →
Data Validation → API Integration → AI Processing → RAG / Knowledge Retrieval →
Business Rules → Database → Notifications → Error Handling → Testing → Documentation

The objective is not simply to learn n8n nodes. It is to develop the ability to: **Understand a business problem → design an automation solution → implement it → integrate AI where appropriate → test it → document it → deliver a reliable working solution.**

Intern Acknowledgment

I have reviewed the 6-Week AI Automation Internship Program and understand the learning objectives, weekly assignments, submission requirements and final project expectations.

Intern Name:	
Signature:	
Date:	
Petalnex Supervisor:	